```
    incolor=gray!30,
    hue=0 1,
    opacity=0.8,
]
\end{pspicture}
\end{document}
```

The last example for 3D plotting is about the representation of some spectroscopic data. In particular, I will consider an absorbance spectrum measured by points from 350 nm to 550 nm with steps of 10 nm, at 8 different times (0, 1, 2, 4, 6, 8, 12, 24 hours). So I have a table 21 rows x 8 columns. The table is present in the preamble as *pgfplotstableread*, the first column is the wavelength and the columns from two to nine are the measured absorbance values. In the preamble I have also added eight different *definecolors*, one for each spectrum, and eight different *pgfplotscreateplotcyclelist* to define some options for each spectrum. The syntax is as follows:

```
\definecolor{col1}{HTML}{F46D43}
...
\definecolor{col8}{HTML}{3288BD}
\pgfplotscreateplotcyclelist{mycol}{%
{draw=col1,line width=1pt,fill=col1,fill opacity=0.6},
...
{draw=col8,line width=1pt,fill=col8,fill opacity=0.6}%
}
```

Each spectrum is then plotted by a loop, as follows:

```
\pgfplotsinvokeforeach{24,12,8,6,4,2,1,0}{
    \addplot3 table[x=nm,y expr=#1,z=abs#1]{\dataabs};
}
```

The result is eight different areas plotted in a 3D space, as shown in Figure 10. For aesthetic reasons, I have added one more row with a z=0 value at 350 nm and at 550 nm. Another option is to plot each spectrum in a simple sequence, not considering the time. In this case, each measured point is specified by its coordinates (x,y,z) and the result is a surface (see Figure 11).

```
\addplot3[
    surf,
    z buffer=sort,
    opacity=0.8,
    shader=flat,
    line width=0.1pt
]
coordinates{
(350,1,0.000)(350,2,0.000)(350,3,0.000) ...
...
... (550,6,0.000)(550,7,0.000)(550,8,0.000)
};
```

Another possibility is to give each spectrum a certain 'thickness' by adding one more 'y' value and then plotting from the last (n=8 or 24 h) to the first (n=1 or 0 h). The resulting plot is a series of ribbons (Figure 12). The contour plot is not directly supported (a Gnuplot installation is required), but it's possible to obtain a pseudo contour

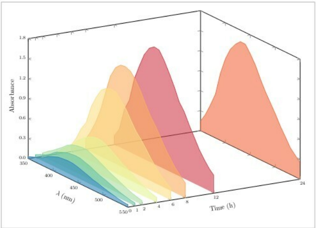

Figure 10: Absorbance spectra 1

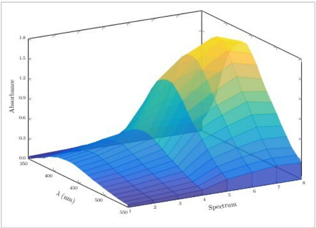

Figure 11: Absorbance spectra 2

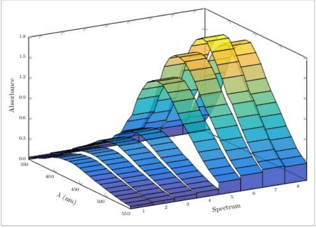

Figure 12: Absorbance spectra 3